

定積分の部分積分法

$$\int_a^b f(x) g'(x) dx = \left[f(x) g(x) \right]_a^b - \int_a^b f'(x) g(x) dx$$

← 積の微分

(例1) 次の定積分を求めよ。

$$\begin{aligned} (1) \quad \int_0^{\frac{\pi}{2}} x \cos x \, dx &= \int_0^{\frac{\pi}{2}} x (\sin x)' dx \\ &= \left[x \sin x \right]_0^{\frac{\pi}{2}} - \int_0^{\frac{\pi}{2}} 1 \cdot \sin x \, dx \\ &= \frac{\pi}{2} - \left[-\cos x \right]_0^{\frac{\pi}{2}} \\ &= \frac{\pi}{2} - 1 \quad // \end{aligned}$$

$$\begin{aligned} (2) \quad \int_1^e x^2 \log x \, dx &= \int_1^e \left(\frac{1}{3} x^3 \right)' \log x \, dx \\ &= \left[\frac{1}{3} x^3 \log x \right]_1^e - \int_1^e \frac{1}{3} x^3 \cdot \frac{1}{x} dx \\ &= \frac{1}{3} e^3 - \frac{1}{3} \left[\frac{1}{3} x^3 \right]_1^e \\ &= \frac{1}{3} e^3 - \frac{1}{9} (e^3 - 1) \\ &= \frac{2e^3 + 1}{9} \quad // \end{aligned}$$

$$\begin{aligned} (3) \quad \int_0^1 x e^{2x} dx &= \int_0^1 x \left(\frac{1}{2} e^{2x} \right)' dx \\ &= \left[\frac{1}{2} x e^{2x} \right]_0^1 - \int_0^1 \frac{1}{2} 1 \cdot e^{2x} dx \\ &= \frac{1}{2} e^2 - \frac{1}{2} \left[\frac{1}{2} e^{2x} \right]_0^1 \\ &= \frac{1}{2} e^2 - \frac{1}{4} (e^2 - 1) \\ &= \frac{e^2 + 1}{4} \quad // \end{aligned}$$

(例2) 次の定積分を求めよ。

$$\begin{aligned} (1) \quad \int_1^2 (x-1)(x-2)^4 dx &= \int_1^2 (x-1) \left(\frac{1}{5} (x-2)^5 \right)' dx \\ &= \left[\frac{1}{5} (x-1)(x-2)^5 \right]_1^2 - \int_1^2 \frac{1}{5} 1 \cdot (x-2)^5 dx \\ &= -\frac{1}{5} \left[\frac{1}{6} (x-2)^6 \right]_1^2 \\ &= \frac{1}{360} \quad // \end{aligned}$$

$$\begin{aligned} (2) \quad \int_\alpha^\beta (x-\alpha)(x-\beta) dx &= \int_\alpha^\beta \left(\frac{1}{2} (x-\alpha)^2 \right)' (x-\beta) dx \\ &= \left[\frac{1}{2} (x-\alpha)^2 (x-\beta) \right]_\alpha^\beta - \int_\alpha^\beta \frac{1}{2} (x-\alpha)^2 \cdot 1 dx \\ &= -\frac{1}{2} \left[\frac{1}{3} (x-\alpha)^3 \right]_\alpha^\beta \\ &= -\frac{1}{6} (\beta-\alpha)^3 \quad // \end{aligned}$$

(例3) 次の定積分を求めよ。

$$\begin{aligned} (1) \quad \int_0^1 x^2 e^{-x} dx &= \int_0^1 x^2 (-e^{-x})' dx \\ &= \left[-x^2 e^{-x} \right]_0^1 + \int_0^1 2x e^{-x} dx \\ &= -e^{-1} + 2 \int_0^1 x (-e^{-x})' dx \\ &= -e^{-1} + 2 \left(\left[-x e^{-x} \right]_0^1 + \int_0^1 1 \cdot e^{-x} dx \right) \\ &= -e^{-1} + 2 \left(-e^{-1} + \left[-e^{-x} \right]_0^1 \right) \\ &= -e^{-1} - 2e^{-1} + 2 \left(-e^{-1} + 1 \right) \\ &= 2 - \frac{5}{e} \quad // \end{aligned}$$

$$\begin{aligned} (2) \quad \int_0^{\frac{\pi}{2}} x^2 \sin 2x \, dx &= \int_0^{\frac{\pi}{2}} x^2 \left(-\frac{1}{2} \cos 2x \right)' dx \\ &= \left[-\frac{1}{2} x^2 \cos 2x \right]_0^{\frac{\pi}{2}} + \frac{1}{2} \int_0^{\frac{\pi}{2}} 2x \cos 2x \, dx \\ &= \frac{\pi^2}{8} + \int_0^{\frac{\pi}{2}} x \left(\frac{1}{2} \sin 2x \right)' dx \\ &= \frac{\pi^2}{8} + \left[\frac{1}{2} x \sin 2x \right]_0^{\frac{\pi}{2}} - \frac{1}{2} \int_0^{\frac{\pi}{2}} 1 \cdot \sin 2x \, dx \\ &= \frac{\pi^2}{8} - \frac{1}{2} \left[-\frac{1}{2} \cos 2x \right]_0^{\frac{\pi}{2}} \\ &= \frac{\pi^2}{8} + \frac{1}{4} (-1 - 1) \\ &= \frac{\pi^2}{8} - \frac{1}{2} \quad // \end{aligned}$$